TARI29 Artificial Intelligence, 7.5 credits. Exam Instructions

Allowed aids:

- VS Code and SWI-Prolog installed at the computers in the exam room
- SWI-Prolog documentation provided as PDF
- The logical equivalence sheet provided as PDF
- Other material provided as PDF

However, you are not allowed to use any other form of assistance. Specifically, it is of course *strictly prohibited* to collaborate with other students or use online resources during the exam.

Instructions for answering questions:

- Each question will provide a task description and the maximum amount of points that can be awarded for a correct answer.
- Type in your answers directly in the Inspera system.
- If a question *mentions* that you can submit a drawing or other type of writing on paper, you can get paper sheets from the personnel in the exam room. Then you need to hand in your sheets to the personnel. The paper sheets will be delivered to the examiners after the exam. If you write on paper, **write in legible (clear) handwriting** so that your answer is well readable.
- For *Prolog problems*, write and debug your code in the VS Code environment. Then copy the Prolog code of your solution from VS Code and paste it directly in the text field of the corresponding question.

Grading:

3	28–41.5 points
4	42–55.5 points
5	56–70 points

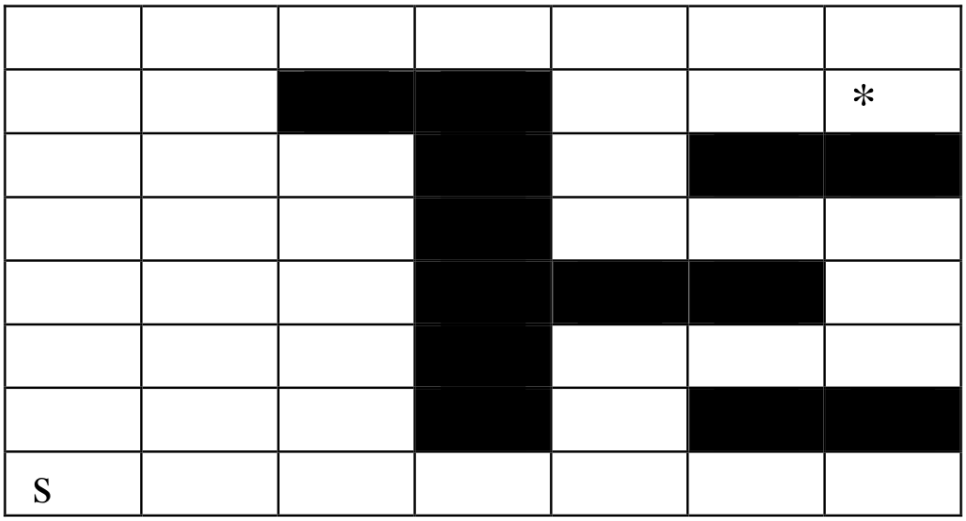
The distribution of points among question sections:

Section	Search	Game Theory	MDP	Logic	Prolog
Points	12	12	9	14	23

Good Luck!

1 Search: A*–Task 1.

Have a look at the very simple world below. The agent is located in the cell marked S and the goal is the cell marked *. The agent can only move horizontal or vertical, i.e., not diagonal, and we want to find the shortest possible path. All black cells are blocked, i.e., cannot be entered by the agent.



We are going to use A* to find the optimal path using the Manhattan distance—that is, disregarding blocked cells—as heuristic function. The Manhattan distance is both admissible and consistent.

- Suggest another heuristic function for this problem that is also admissible. Is this alternative more or less suitable for solving this problem? Give a justification for your answer.

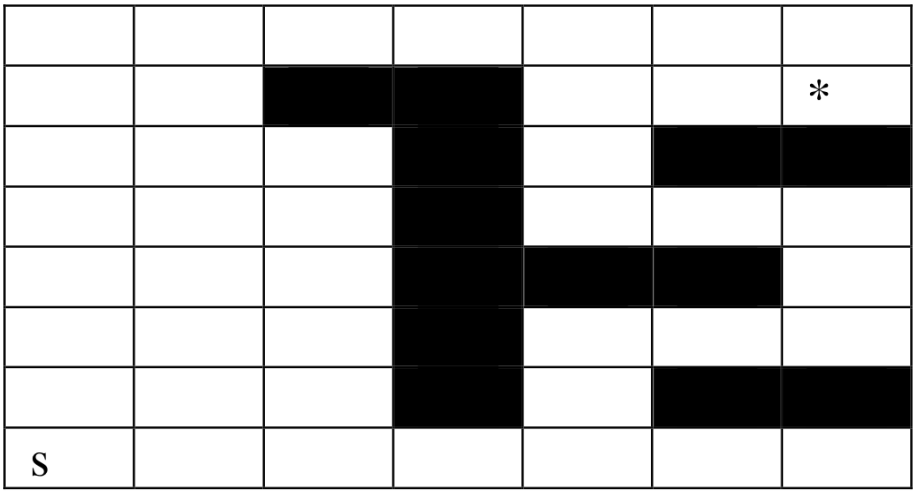
You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

2 Search: A*–Task 2.

Have a look at the very simple world below. The agent is located in the cell marked *S* and the goal is the cell marked *. The agent can only move horizontal or vertical, i.e., not diagonal, and we want to find the shortest possible path. All black cells are blocked, i.e., cannot be entered by the agent.



We are going to use A* to find the optimal path using the Manhattan distance—that is, disregarding blocked cells—as heuristic function. The Manhattan distance is both admissible and consistent.

- Formally show that the Manhattan distance is consistent. That is, show that for each state *n* and each successor state *n'*:
  $h(n) \leq c(n, n') + h(n')$
  where *h* is the heuristic function, and $c(n, n')$ is the cost of reaching *n'* from *n*.

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

3 Search: A*–Task 3.

Have a look at the very simple world below. The agent is located in the cell marked S and the goal is the cell marked *. The agent can only move horizontal or vertical, i.e., not diagonal, and we want to find the shortest possible path. All black cells are blocked, i.e., cannot be entered by the agent.

						*
S						

We are going to use A* to find the optimal path using the Manhattan distance—that is, disregarding blocked cells—as heuristic function. The Manhattan distance is both admissible and consistent.

- Show the order in which the states are expanded by writing numbers in the cells, starting with 0 in position S and 1 in the next expanded state, etc. Please note the following rules: if there are several states with equal values, the agent first expands the last state found. (The frontier is a stack.) If this is not enough to break the tie, the order is E, S, W, N.

You may use paper sheets to answer this question.

If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

4 Game Theory - Task 1

Consider the payoff matrix below for a game in normal form.

		Player B		
		X	Y	Z
Player A	X	5,5	3,0	0,2
	Y	4,1	2,1	1,0
	Z	0,0	2,4	4,5

Are there any pure Nash Equilibria or Pareto Optimal outcomes in the game? If so, what are they (and why)?

You may use paper sheets to answer this question.
Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 5

5 Game Theory - Task 2

Let us assume the following scenario. While SAAB is an established car manufacturer, a new company has emerged, Nordicars AB, increasing its share in the market. Both companies want to develop a new Sedan-type car. They can either team up and aim to develop a new Sedan-type car, or each can work independently to develop a less ambitious car. Based on their decision, the following cases may occur:

- if they team up and agree to develop the new Sedan-type car together, then a better product will be released on the market, leading to profits of 2 billion SEK for each company
- if they defect and work on their own project, they will develop a less ambitious car, leading to lower profits for both, equal to 1 billion SEK
- but if one cooperates (i.e., starts working on an agreed common project) and the other defects (i.e., even though they have agreed on a common project, this company starts working on their own car), then the common project will fail (i.e., no profits) while the latter develops a less ambitious car and earns a profit of 1 billion SEK.

Represent the above situation as a game in its Normal Form. What is the dominant strategy of each company and why? Can we predict what would be the outcome of this game? Do we make any assumptions to make this prediction?

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

6 Game Theory - Task 3

Calculate a Nash Equilibrium with mixed strategies from the payoff matrix below and state the value of the game at this equilibrium. Use the method of assuming the opponent should be indifferent between its actions. Show all your calculations.

		Player B	
		X	Y
Player A	X	1,1	3,0
	Y	4,1	2,2

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

7 MDP / Reinforcement Learning - Task 1

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, a *discount factor* $\gamma = 1/4$, a *reward function* R and a *transition function* T as shown below:

$$R(s,a,s') = \begin{cases} 1/2 & \text{if } s' = S_1 \\ -1/2 & \text{if } s' = S_2 \end{cases}$$

s	a	s'	T(s,a,s')
S_1	A_1	S_1	$3/4$
S_1	A_1	S_2	$1/4$
S_1	A_2	S_1	$1/2$
S_1	A_2	S_2	$1/2$
S_2	A_1	S_1	$1/3$
S_2	A_1	S_2	$2/3$
S_2	A_2	S_1	$1/2$
S_2	A_2	S_2	$1/2$

Perform one iteration of *value iteration* and enter the *value* for $V_1(S_1)$ in the table below (you don't have to enter a value into the gray cell). Use the initial values for $V_0(s)$ below instead of initializing all values to 0. Show all your calculations.

s	$V_0(s)$	$V_1(s)$
S_1	4	?
S_2	4	

$$V_{k+1}(s) \leftarrow \max_a \sum_{s'} T(s,a,s')[R(s,a,s') + \gamma V_k(s')]$$

You may use paper sheets to answer this question.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 3

8 MDP / Reinforcement Learning - Task 2

Assume an MDP consists of the two states S_1 and S_2, the two actions A_1 and A_2, a *transition function T* as shown below:

s	a	s'	$T(s,a,s')$
S_1	A_1	S_1	$1/4$
S_1	A_1	S_2	$3/4$
S_1	A_2	S_1	$1/2$
S_1	A_2	S_2	$1/2$
S_2	A_1	S_1	$1/4$
S_2	A_1	S_2	$3/4$
S_2	A_2	S_1	$1/2$
S_2	A_2	S_2	$1/2$

Assume *Q-learning* with a *learning* rate $\alpha = 1/4$ and without discount (i.e., $\gamma = 1$) is used. The following episode is observed (i.e. first the agent executes the action A_1 in state S_1, ending up in state S_1 with the reward *-1*, followed by the agent executing the action A_2 in state S_1, ending up in state S_1 with the reward *1*):

s_1	a_1	r_1	s_2	a_2	r_2	s_3
S_1	A_1	-1	S_1	A_2	1	S_2

Use the initial *Q-values* Q_0 in the table below, and enter the values for Q_1 and Q_2. Show all your calculations.

s	a	$Q_0(s,a)$	$Q_1(s,a)$	$Q_2(s,a)$
S_1	A_1	$1/2$	?	?
S_1	A_2	$1/8$	?	?
S_2	A_1	$5/4$	?	?
S_2	A_2	$1/4$	?	?

$$Q_{k+1}(s,a) \leftarrow Q_k(s,a) + \alpha[r + \gamma \max_{a'} Q_k(s',a') - Q_k(s,a)]$$

You may use paper sheets to answer this question.

Fill in your answer here.If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

9 Logic: First-order logic—Task 1

Consider the following knowledge base, given in natural language:

```
Harry is a wizard.
All evil wizard smoke.
Harry knows a wizard who smokes.
Dumbledore is the only wizard who smokes and who is not evil.
```

Translate the knowledge base into first-order logic, using one FOL sentence for each sentence above.

You must use the following predicates and constants:

- *Unary predicates*; i.e. taking one argument: wizard, smokes, evil.
- *Binary predicates*; i.e. taking two arguments: knows.
- *Constants*; i.e. names of people: Harry, Dumbledore

You may use paper sheets to answer this question.

Alternatively, you can use the following way to quickly type in logical formulae in Inspera.

- For logical connectives, use symbols that are used in the programming languages Java, C\#, and C++: '!' for negation, '&&' for conjunction, '||' for disjunction, '->' for implication , '' <--> for biconditional
- For universal quantifier use 'forall', and for existential quantifier use 'exists'.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

10 Logic: First-order logic—Task 2

Consider the following knowledge base consisting of one first-order logic sentence:

$$\exists x.\,(\mathbf{wizard}(x) \wedge \forall y.\,(\mathbf{wizard}(y) \Rightarrow \mathbf{knows}(x, y)))$$

Define two first-order models, one for which this sentence is true and one for which this sentence is false. Formally specify the domain and an interpretation function relative to this domain for each model.

You may use paper sheets to answer this question.

Alternatively, you can use the following way to quickly type in logical formulae in Inspera.

- For logical connectives, use symbols that are used in the programming languages Java, C\#, and C++: '!' for negation, '&&' for conjunction, '||' for disjunction, '->' for implication , '' for biconditional
- For universal quantifier use 'forall', and for existential quantifier use 'exists'.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 4

11 Logic: First-order logic—Task 3

Consider the following knowledge base consisting of one first-order logic sentence:

$$\exists x.\,(\mathbf{wizard}(x) \wedge \forall y.\,(\mathbf{wizard}(y) \Rightarrow \mathbf{knows}(x, y)))$$

Use resolution to proof that this knowledge base entails the query sentence

$$\exists z.\ \mathbf{knows}(z, z)$$

This proof requires you to convert the knowledge base and the negated query into a set of clauses. Recall that the steps for doing this are the following:

1. Eliminate biconditionals and implications;
2. Move negation inwards, using the logical equivalences;
3. Standardize variables if necessary: each quantifier should use a different variable;
4. Drop existential quantifiers and replace existential variables by Skolem functions of the enclosing universally quantified variables;
5. Drop universal quantifiers;
6. Use distributivity law to distribute disjunction over conjunction.

You may use paper sheets to answer this question.
Alternatively, you can use the following way to quickly type in logical formulae in Inspera.

- For logical connectives, use symbols that are used in the programming languages Java, C\#, and C++: '!' for negation, '&&' for conjunction, '||' for disjunction, '->' for implication , '' for biconditional
- For universal quantifier use 'forall', and for existential quantifier use 'exists'.

Fill in your answer here. If you answered on paper, type in here "Answered on paper"

Maximum marks: 6

12 Prolog: Constraint Logic Programming (CLP)

Define the predicate **puzzle(Solution)** that solves the cryptarithmetic puzzle below using CLP(FD). You need to find all concrete solutions.

```
B A T
+ H I T
+ I N N I N G
-------------
 B A T T E R
```

Use separate lists to represent each word, so that after running the query **?-puzzle(Solution)**, each word in a solution of the puzzle would be represented by a list of digits. Like in the example given in the lecture:

```
Solution = ([9, 5, 6, 7] +[1, 0, 8, 5]=[1, 0, 6, 5, 2])).
```

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 8

13 Prolog: Lists

Define the procedure **split(Numbers, Bigger, Smaller)** which splits a list of numbers into two lists: bigger or equal to ‘5’ (including five) and smaller than five.

For example, this query

```
?- split([6, -1, 0, 5, -2, 3], [6, 5], [-1, 0, -2, 3]).
```

would evaluate to *true*.

You have to use *cut* (‘!’) in your definition. You may <u>not</u> use any built-ins (library predicates are <u>not</u> allowed) in this task.

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 7

14 Prolog: Frames

(*4 points*). Create three frames that will represent the following information:

- COMP is a department. Its full name is Department of Computing. COMP is part of the JTH school. COMP teaches two programmes: AIE and CSE.
- AIE is an educational programme. Its full name is AI Engineering. AIE includes courses on Artificial Intelligence, Knowledge Representation, and Machine Learning.
- CSE is an educational programme. Its full name is Computer Science and Engineering. CSE includes courses on Data Structures, Algorithms, and Databases.

(*4 points*). Now, define the predicate **all_programmes/0** that will output all the courses that are taught by the Department of Computing when the user presses ';' or space bar. It is enough to write course names as constants. An example output is provided below.
Hint: You may need to use the built-in predicate **`=..`** ('univ') to solve this problem.

```
?- all_programmes.
artificial_ntelligence
true ;
knowledge_representation
true ;
machine_learning
true ;
data_structures
true ;
algorithms
true ;
databases
true.
```

Copy the Prolog code of your solution from VS Code and paste it directly in this text field.

Maximum marks: 8